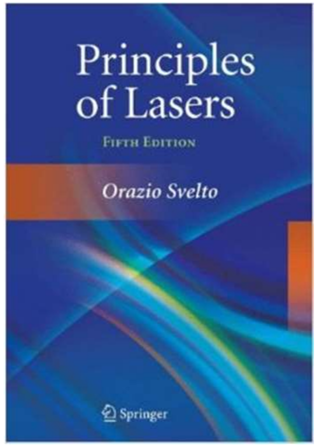
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers



Ch4-2
Oscillation Characteristics of Lasers

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Chapter 4: Oscillation Characteristics

❖ 4.2 Oscillation Mode (Output Mode)

- Mode competition in lasers with homogenous broadening
- Multi-longitudinal mode oscillation of lasers with inhomogeneous broadening

❖ 4.3 Output Optical Power and Energy

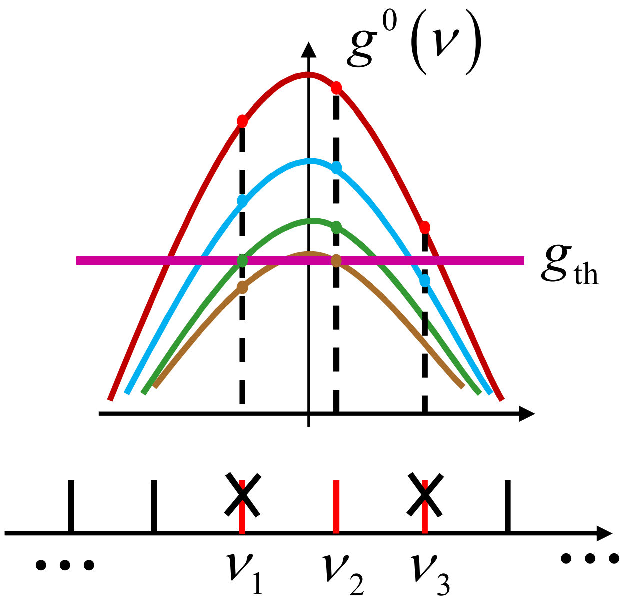
- Output power for CW or long-pulse laser
- Output power of short-pulse lasers（$t_0 << \tau_s$)

4.2 Oscillation mode (output mode)

Discuss the output modes of lasers from the mechanism of gain saturation

❖ Mode competition in lasers with homogenous broadening

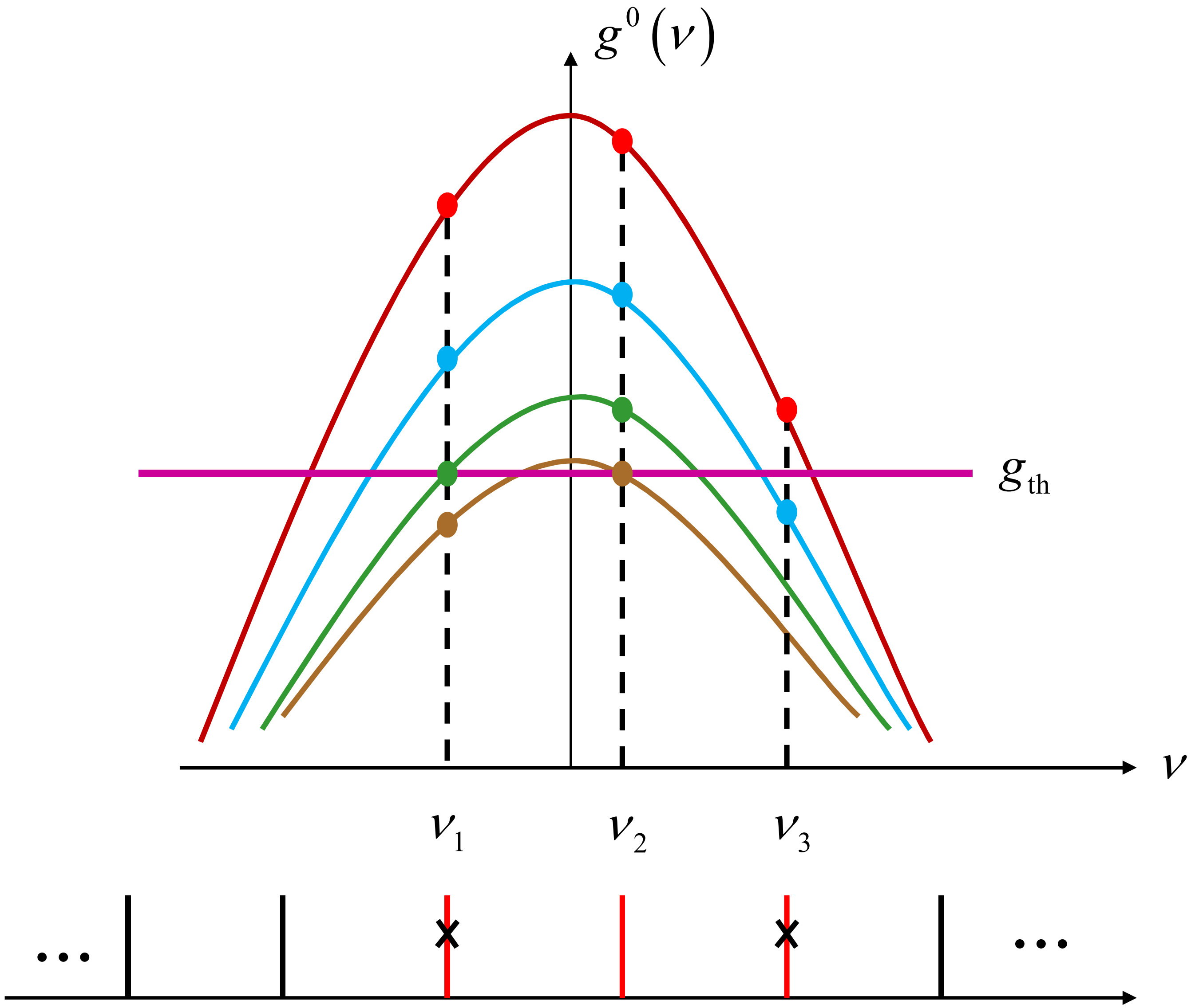
□ Self-selection mode due to homogenous saturation of gain



Gain Curve	ν_1		ν_2		ν_3	
$g(\nu)$	$g(\nu_1)$	I_1	$g(\nu_2)$	I_2	$g(\nu_3)$	I_3
Red	$> g_{th}$	↑	$> g_{th}$	↑	$> g_{th}$	↑
Blue	$> g_{th}$	↑	$> g_{th}$	↑	$= g_{th}$	-
Green	$= g_{th}$	-	$> g_{th}$	↑	$< g_{th}$	0
Brown	$< g_{th}$	0	$= g_{th}$	-	$< g_{th}$	0

steady-state gain

Conclusion: the output of a homogeneous broaden steady-state laser should be a single longitudinal mode. Its working frequency locates at central frequency of the gain curve. Other longitudinal modes will be suppressed.


$g^0(\nu)$
g_{th}
ν
ν_1
ν_2
ν_3

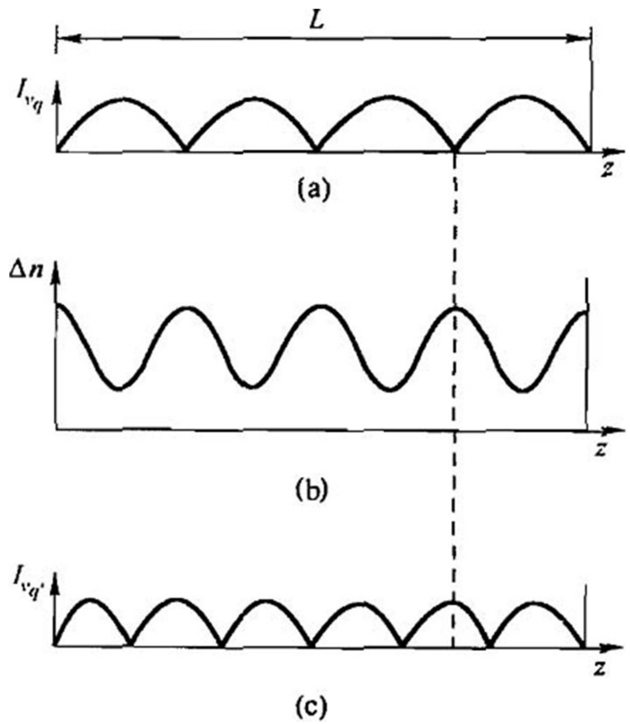
☐ Multimode oscillation induced by spatial hole-burning

1) Spatial hole burning effect

- Assume homogeneous transverse distribution, only consider *z*-direction distribution
- Standing wave ➡ gain distribution $g(z)$ ➡ spatial hole-burning

2) Physical mechanism

- Different longitudinal modes can use the activated atoms in different positions inside the cavity



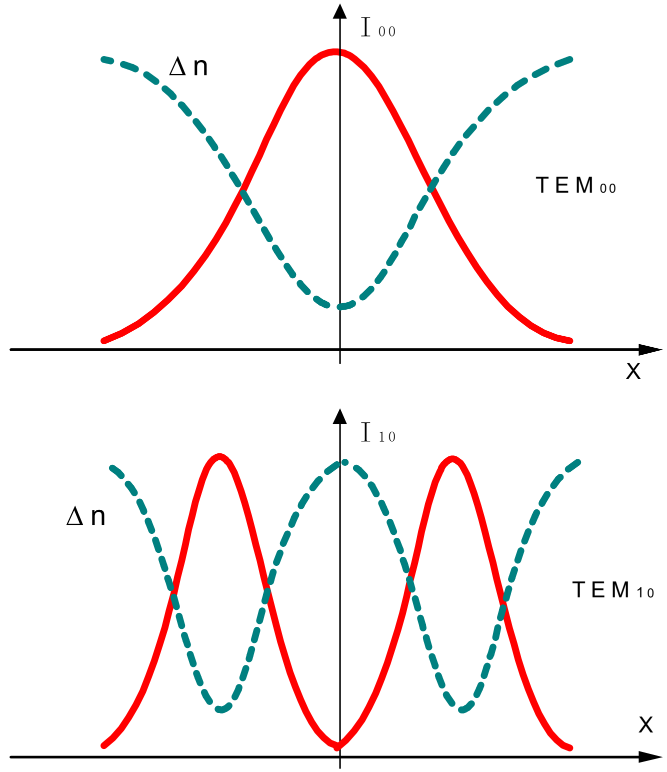
3) Formation conditions

- Standing wave cavity: spacing between burning holes is at the scale of wavelength
- Slow spatial transfer speed of activated atoms

- Gas: random thermal motion, fast spatial transfer speed, difficult to form spatial hole-burning.
- Solid: example, Cr ions bound to the lattice structure, $\lambda/4$ spatial transfer needs 10^{-4}s
- Semiconductor: 10^{-7}s

4) Formation mechanism of transverse hole-burning

- Nonuniform distribution of population of transverse modes, transverse hole-burning size is large (mm scale), spatial transfer cannot eliminate the nonuniformity.
- With enough pumping, different transverse modes can utilize the activated atoms in different positions and hence form multi-transverse modes oscillation.

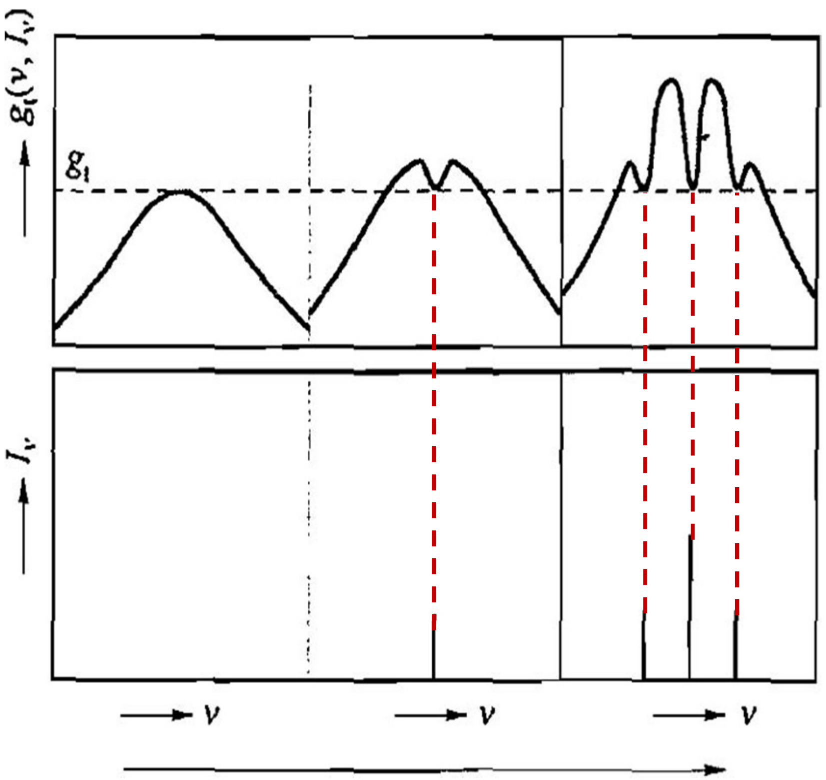


How about for pulse excitation? Mode competition time is in the range of μs

❖ Multi-longitudinal mode oscillation of lasers with inhomogeneous broadening

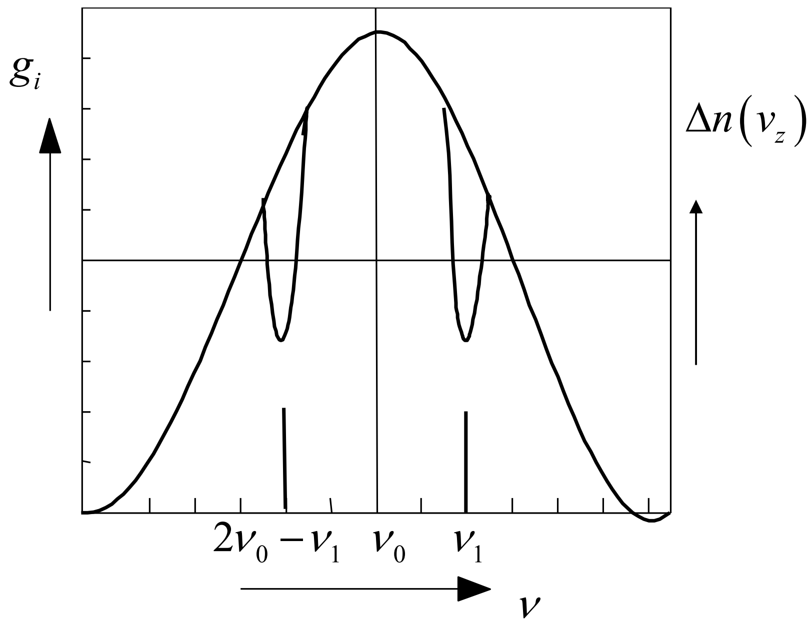
External pump↑ → g^0 ↑→ longitudinal modes that meet threshold conditions ↑ → mode numbers ↑

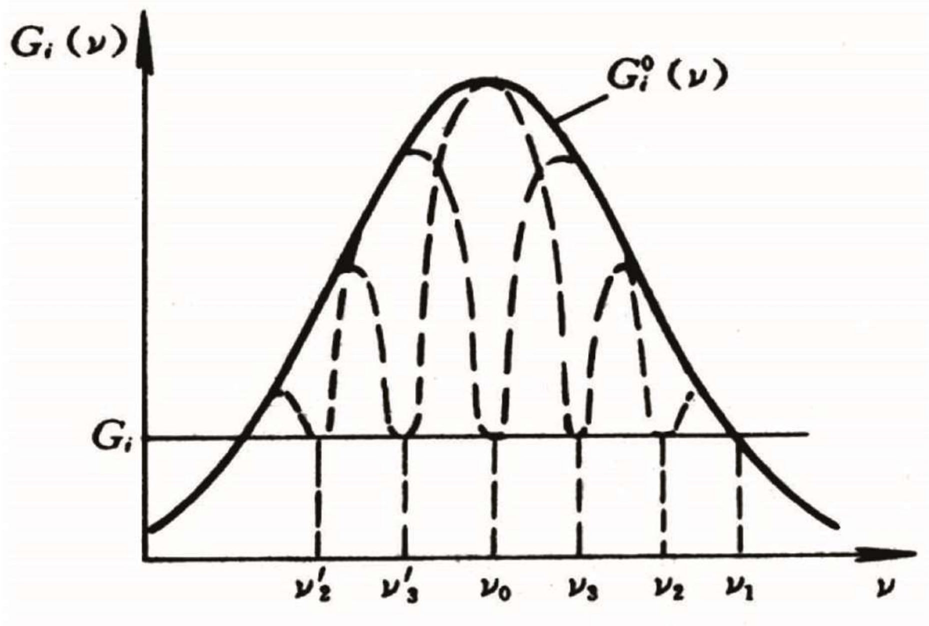
- For incident light (ν, I_ν), gain saturation will only happen in the frequency range of ν+dν. Laser oscillation can still happen for other frequencies.
- For the case of Doppler broadening, symmetrical hole burning at both sides of central frequency ν_0. If ν_q=ν_0, modes ν_{q+1} and $\nu_{q\text{-}1}$ will compete, and result in random fluctuation of their output power
- If frequency spacing between ν_q and ν_{q+1} is less than the spacing of burning hole width, their burning hoes will overlap and result in mode competition



□ Mode competition in inhomogeneous broadened lasers with standing wave

- Longitudinal mode frequencies ν_1 and ν_2 symmetrically distribute on both sides of the central frequency ν_0, and deplete the inverted population
- The spatial burning holes will overlap for adjacent longitudinal modes if the spacing is small enough



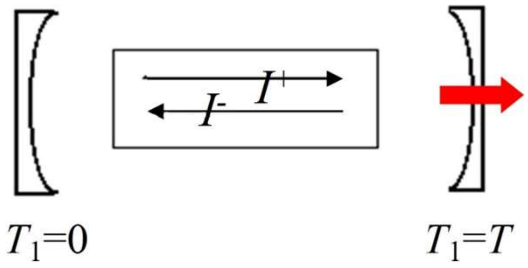


4.3 Output Optical Power and Energy

Discuss the average intensity at steady-state, estimate the output power

Area of the laser beam's cross-section

$$P = I^{+}TA \;\Rightarrow\; P = \frac{1}{2}ITA \;\Rightarrow\; P_{out} \Leftrightarrow I^{+}$$



❖ Output power for CW or long-pulsed laser

☐ Single-mode laser (Assume mode l with frequency of ν_q)

$$\begin{cases}\Delta n^{0} > \Delta n_{\mathrm{t}} \\ g^{0}\left(\nu_q\right) > g_{\mathrm{t}}\end{cases} \Rightarrow \begin{cases}N_l \uparrow \\ I_{\nu_q} \uparrow\end{cases} \xRightarrow{\text{saturation}} \begin{cases}\Delta n \downarrow \\ g\left(\nu_q, I_{\nu_q}\right)\downarrow\end{cases} \xRightarrow{g(\nu)>g_{\mathrm{t}}} \begin{cases}N_l \uparrow \\ I_{\nu_q} \uparrow\end{cases} \xRightarrow{\text{enhanced saturation}} \begin{matrix}\Delta n \downarrow \\ g\left(\nu_q, I_{\nu_q}\right)\downarrow\end{matrix} \Rightarrow$$

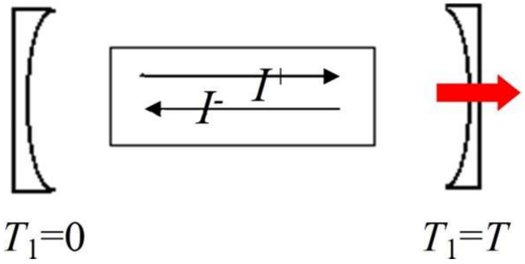
$$\boxed{\begin{cases}g\left(\nu_q, I_{\nu_q}\right) = g_{\mathrm{t}} = \alpha \\ \dfrac{dN_l}{dt} = 0, \quad \dfrac{d\Delta n}{dt} = 0\end{cases} \Rightarrow \begin{matrix} I_{\nu_q} & \text{fixed intensity} \\ N_l & \text{no change}\end{matrix}}$$

$$N_l \upsilon h\nu = I_{\nu_q}^{+} + I_{\nu_q}^{-} \Rightarrow I_{\nu_q}^{+} = \frac{1}{2}N_l \upsilon h\nu$$

$$\boxed{P = I_{\nu}TA = \frac{1}{2}N_l h\nu\upsilon AT}$$

Lasers with homogeneous broaden ($\nu=\nu_0$)

Assume $T<<1$, $\quad I_\nu^+ \approx I_\nu^-, \quad I = I_\nu^+ + I_\nu^- \approx 2I_\nu$



I_ν^+, I_ν^- participate saturation simultaneously

How to calculate I_ν inside cavity

$$g_H(\nu, I_\nu) = \frac{g_m}{1+(2I_\nu/I_s)} = \frac{\delta}{l} = g_t$$

$$I_\nu = \frac{1}{2}I_s\left(\frac{g_m l}{\delta}-1\right) \Rightarrow \boxed{P = \frac{1}{2}I_s TA\left(\frac{g_m l}{\delta}-1\right)}$$

Gas lasers: $T<<1 \qquad 2\delta \approx T+\alpha \qquad \alpha <<1$ $\qquad \boxed{P = \frac{1}{2}I_s TA\left(\frac{2g_m l}{\alpha+T}-1\right)}$

- α — single-loop net loss factor without considering reflection
- A — effective sectional area of the laser beam

Discussion 1: Output power ~ pumping power ($P_\mathrm{p}\uparrow \rightarrow P\uparrow$)

- Solid-state lasers (optical pumping)

$$\boxed{P=\frac{1}{2}I_sTA\left(\frac{g_m l}{\delta}-1\right)}$$

$$g_m=\Delta n^0\sigma_{21} \qquad P_\mathrm{p}=\frac{h\nu_\mathrm{p}\Delta n^0 V}{\eta_F\tau_s}$$

$$g_\mathrm{t}=\Delta n_\mathrm{t}\sigma_{21} \qquad P_\mathrm{pt}=\frac{h\nu_\mathrm{p}\Delta n_\mathrm{t} V}{\eta_F\tau_s}$$

$$\frac{g_m}{g_\mathrm{t}}=\frac{P_\mathrm{p}}{P_\mathrm{pt}}=\frac{g_m l}{\delta}$$

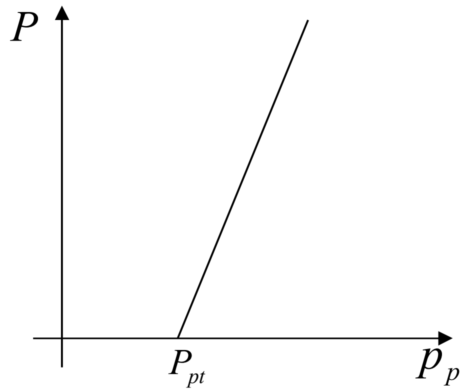
$$\eta_2=\frac{\tau_2}{\tau_s} \qquad \eta_F=\eta_1\eta_2$$

$$\frac{\Delta n_\mathrm{t}}{\tau_2}=\frac{P_\mathrm{pt}\eta_1}{h\nu_\mathrm{p}V}=\frac{P_\mathrm{pt}\eta_1}{h\nu_\mathrm{p}Sl}$$

$$P=\frac{1}{2}I_sTA\left(\frac{g_m l}{\delta}-1\right)=\frac{1}{2}h\nu_0\frac{\Delta n_\mathrm{t} l}{\delta\tau_2}TA\left(\frac{P_\mathrm{p}}{P_\mathrm{pt}}-1\right)=\frac{A}{S}\frac{\nu_0}{\nu_\mathrm{p}}\eta_0\eta_1P_\mathrm{pt}\left(\frac{P_\mathrm{p}}{P_\mathrm{pt}}-1\right)$$

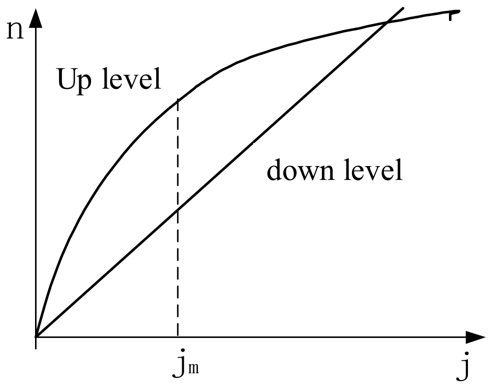
$$I_s=\frac{h\nu_0}{\sigma_{21}\tau_2}, \quad \frac{\delta}{l}=\Delta n_\mathrm{t}\sigma_{21} \qquad\qquad \eta_0=T/2\delta$$

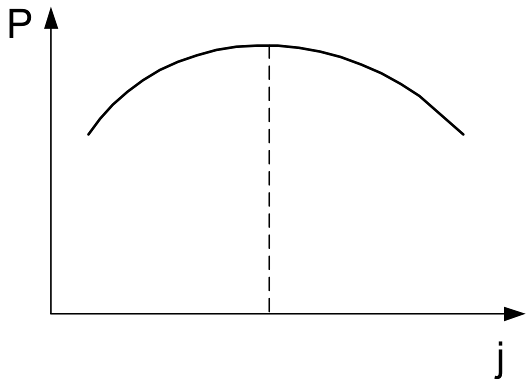
When $P_\mathrm{p}>P_\mathrm{pt}$, P increases linearly with P_p



- The output power of the optically pumped laser is converted from the pump power beyond P_pt
- I_ν is determined by the gain beyond g_t

- Gas lasers: exist optimum discharge current (P229)





Empirical formula of the optimal discharge current for He-Ne laser

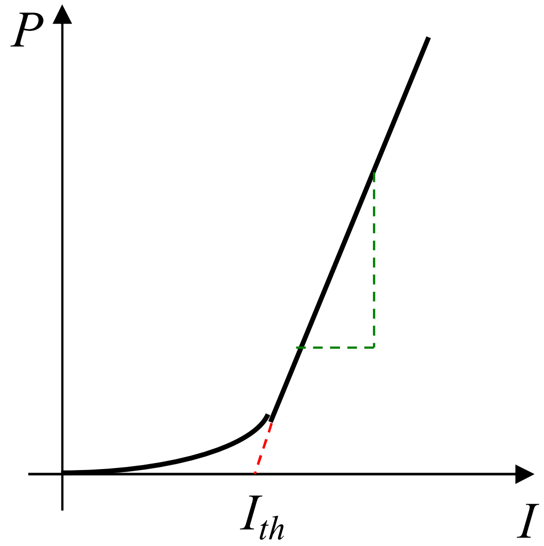
$$g_m = 1.4 \times 10^{-2} \frac{l}{d}$$

- Semiconductor lasers (P278)

$$P = \frac{J - J_{th}}{e} \eta_i \hbar\omega \qquad P_{out} = \frac{\ln(1/r)}{\alpha L + \ln(1/r)} P$$

$$\eta_D = \frac{P_{out}}{P_{in}} = \frac{J - J_{th}}{J} \eta_i \frac{\hbar\omega}{eV} \frac{\ln(1/r)}{\alpha L + \ln(1/r)}$$

- η_i – Carrier recombination probability
- η_D – LD (laser diode) efficiency
- $P_{in} = JV$



Discussion 2: Output power ~ transmittance (optimum T)

- Can be Experimentally measured and calculated

$$T\uparrow \qquad \begin{matrix} P_{out}\uparrow \\ \delta\uparrow\rightarrow g_t\uparrow\rightarrow I_v\downarrow \end{matrix} \quad \Longrightarrow \quad T_m \quad \boxed{P=\frac{1}{2}I_sTA\left(\frac{g_m l}{\delta}-1\right)=\frac{1}{2}I_sTA\left(\frac{2g_m l}{\alpha+T}-1\right)}$$

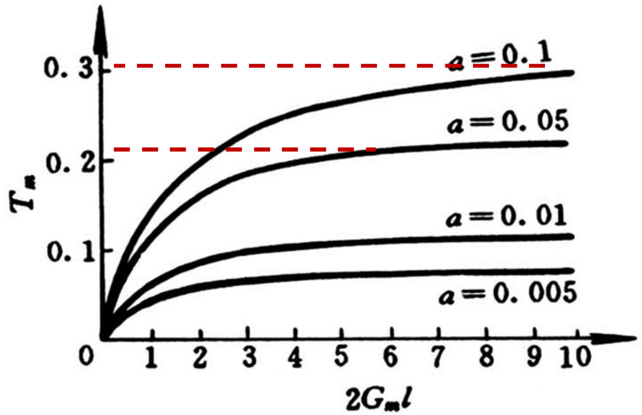
T_m: 1) Experimentally fixed P_p, change T and measure P_{out}

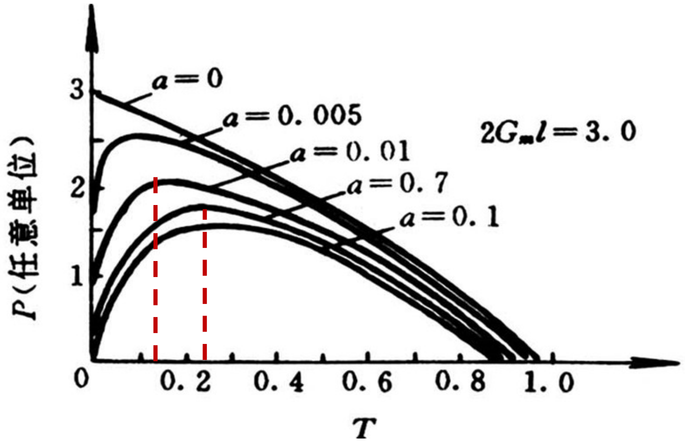
2) Calculate $dP/dT=0$

$$P=\frac{1}{2}I_sTA\left(\frac{2g_m l}{a+T}-1\right) \Longrightarrow \frac{dP}{dT}=AI_s g_m l\frac{a}{(a+T)^2}-\frac{1}{2}AI_s \xrightarrow{dP/dT=0}$$

$$(a+T)^2=2g_m la \Longrightarrow \boxed{T_m=\sqrt{2g_m la}-a} \Longrightarrow \boxed{P_m=\frac{1}{2}AI_s\left(\sqrt{2g_m l}-\sqrt{a}\right)^2}$$

- When T<<1, $2\delta\approx T+a$, a is the single-loop net loss factor





Discussion 3: Output power ~ length of gain material ($l\uparrow \rightarrow P\uparrow$)

Discussion 4: Output power ~ property of gain material $\boxed{P=\frac{1}{2}I_sTA\left(\frac{g_m l}{\delta}-1\right)}$

four-level system

$$\left\{\begin{aligned} P&=\frac{1}{2}I_sTA\left(\frac{g_m l}{\delta}-1\right)\\ I_s&=h\nu_0/\sigma_{21}\tau_2\\ g_m&=\left(nW_{03}\tau_2\right)\sigma_{21}\end{aligned}\right\} \Rightarrow P=\frac{1}{2}TAh\nu_0\left(nW_{03}\frac{l}{\delta}-\frac{1}{\sigma_{21}\tau_2}\right)$$

$$W_{03}\uparrow, n\uparrow, \sigma_{21}\uparrow, \tau_2\uparrow \rightarrow P\uparrow$$

three-level system

$$\left\{\begin{aligned} I_s&=\frac{h\nu_0}{2\tau_{21}}\left(\frac{1}{\tau_2}+W_{13}\right) \qquad \Delta n_0=\frac{n\left(W_{13}-1/\tau_2\right)}{W_{13}+1/\tau_2}\\ g_m&=\frac{n\left(W_{13}-1/\tau_2\right)}{W_{13}+1/\tau_2}\sigma_{21} \qquad \tau_2=\frac{1}{A_{21}+S_{21}}\\ P&=\frac{1}{4}ATh\nu_0\left[W_{13}\left(\frac{nl}{\delta}-\frac{1}{2\sigma_{21}}\right)-\frac{1}{\tau_2}\left(n\frac{l}{\delta}+\frac{1}{\sigma_{21}\tau_2}\right)\right]\\ &W_{13}\uparrow, n\uparrow, \sigma_{21}\uparrow, \tau_2\uparrow \rightarrow P\uparrow\end{aligned}\right.$$

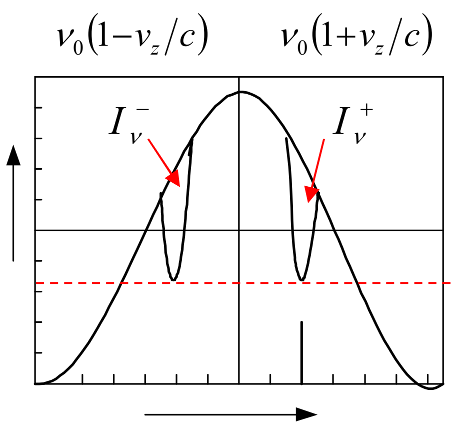
$W_{13}>1/\tau_2$
to have $\Delta n^0>0$

Note for this physical model 1) assume T<<1; 2) $I_\nu^+=I_\nu^-$

□ Lasers with inhomogeneous broaden (single-mode)

When $\nu_q \neq \nu_0$, I_ν^+ and I_ν^- will have their own burning-holes

$$g_i\left(\nu_q, I_{\nu_q}\right) = \frac{g_i^0\left(\nu_q\right)}{\sqrt{1+\left(\frac{I_\nu^+}{I_s}\right)}} = \frac{\delta}{l} \Rightarrow I_\nu^+ = I_s\left[\left(\frac{g_i^0\left(\nu_q\right)l}{\delta}\right)^2 - 1\right]$$



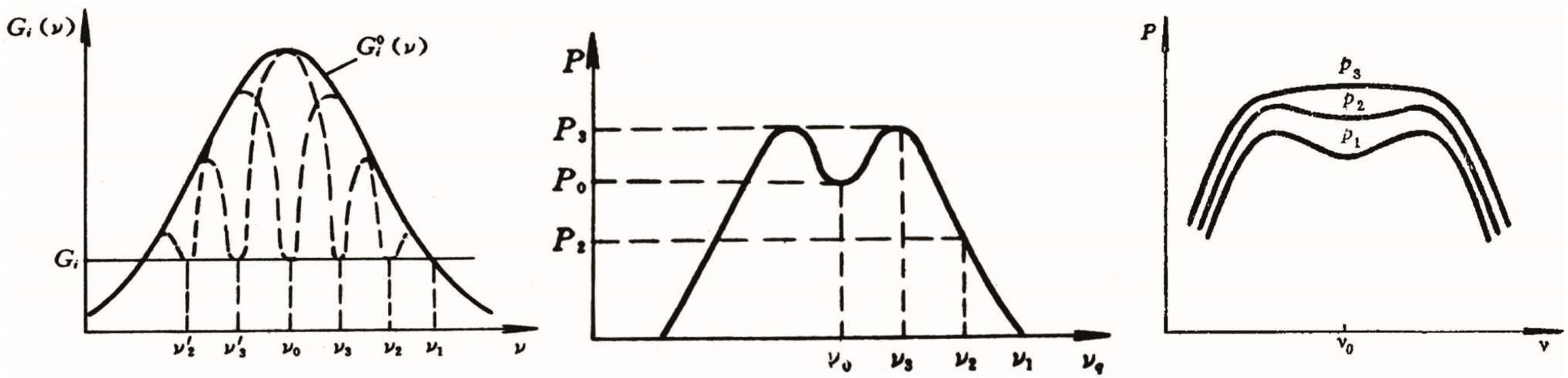
$$\Rightarrow \quad I_\nu^+ = I_s\left\{\left[\frac{g_m l}{\delta}\exp\left(-4\ln 2\frac{\left(\nu_q-\nu_0\right)^2}{\Delta\nu_D^{\,2}}\right)\right]^2 - 1\right\}$$

$$P = AI_\nu^+T = AI_sT\left[\left(\frac{g_i^0\left(\nu_q\right)l}{\delta}\right)^2 - 1\right]$$

$$P = AI_\nu^+T = \frac{1}{2}AI_sT\left[\left(\frac{g_m l}{\delta}\right)^2 - 1\right]$$

When $\nu_q = \nu_0$ $\quad I_\nu = I_{\nu_0}^+ + I_{\nu_0}^- \approx 2I_{\nu_0}^+$ similarly we have

Lamb Dip: Relation between single mode output power and frequency



- $P \propto$ area of the burning-hole (characterize the inverted populations which contribute to the lasing)

a) Burning-hole overlap condition $\left|\nu_q - \nu_0\right| < \dfrac{\Delta\nu}{2}\sqrt{1+\dfrac{I_{\nu_q}}{I_s}}$

b) Width of the Lamb dip (δ_ν) ≈ Hole width $\delta\nu = \Delta\nu_H\sqrt{1+\dfrac{I_{\nu_q}}{I_s}}$

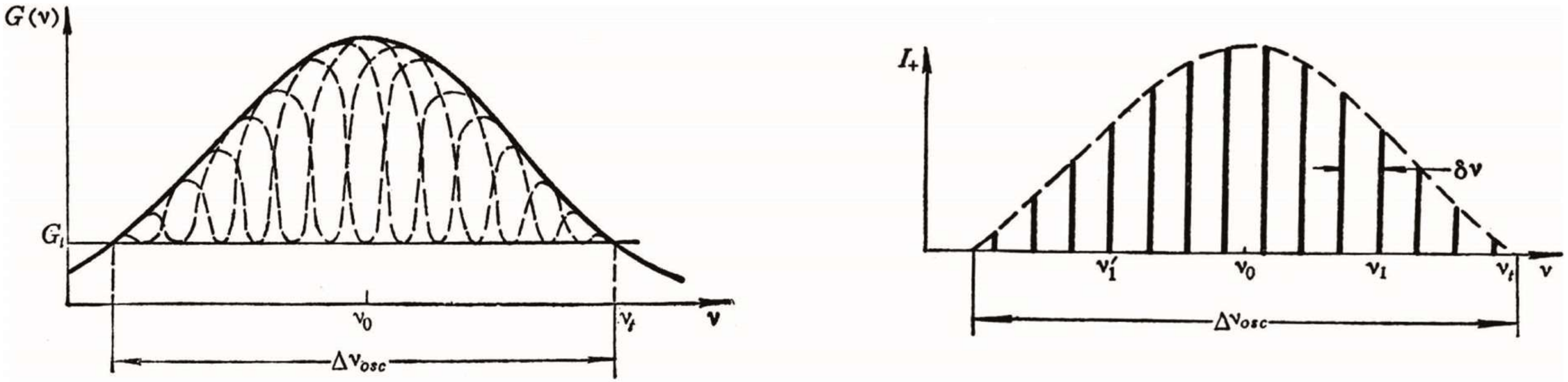
c) Width of the Lamb dip (δ_ν) ~ $\Delta\nu_{\mathrm{L}}$

Pressure↑→ Collision broadening $\Delta\nu_L$↑→ Lamb dip width δ_ν↑, shallow depth

□ Multi-mode lasers

➢ Inhomogeneous broadening:

$\Delta\nu_q$ large enough, NO overlapping of burning holes, calculate the output power for each longitudinal mode, their sum will be the total output power.



➢ Homogeneous broadening: (solid-state lasers)

P_{out} must be calculated using the multi-mode rate equation with simplified assumption (same loss for each mode, rectangular linear function)

Area of the laser beam's cross-section

$$P = \frac{A}{S}\frac{\nu_0}{\nu_\mathrm{p}}\eta_0\eta_1 P_\mathrm{pt}\left(\frac{P_\mathrm{p}}{P_\mathrm{pt}} - 1\right)$$

Area of the gain material's cross-section

❖ Output power of short-pulse lasers（$t_0 << \tau_s$)

Four-level system:

output energy→ stimulated photon number → n_2 → absorbed pumping energy

$E_0 \to E_3$		$E_3 \to E_2$		$E_2 \to E_1$		E_{in}
Absorb pumping energy	→	Activated photon number	→	Stimulated photon number	→	Intra-cavity laser energy
E_p		$\dfrac{E_p \eta_1}{h\nu_p} > \Delta n_t V$		$\left(\dfrac{E_p \eta_1}{h\nu_p} - n_{2t} V\right)$		$h\nu_0 \left(\dfrac{E_p \eta_1}{h\nu_p} - n_{2t} V\right) \dfrac{A}{S}$

η_1 – Pumping efficiency

V – Volume

$$E_{pt} = \frac{h\nu_p \Delta n_t V}{\eta_1}$$

→ Output energy (E)

$$\frac{A}{S} \frac{\nu_0}{\nu_p} \frac{T}{T+a} \eta_1 E_{pt} \left(\frac{E_p}{E_{pt}} - 1 \right) \xrightarrow{\frac{T}{T+a} = \eta_0} \boxed{E = \frac{A}{S} \frac{\nu_0}{\nu_p} \eta_0 \eta_1 E_{pt} \left(\frac{E_p}{E_{pt}} - 1 \right)}$$